

ActInf GuestStream 053.1 ~ A case for chaos theory inclusion in neuropsychanalytic modeling

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hello and welcome everyone this is guest stream number 53.1 at the Active inference Institute it's August 23rd 2023 we're going to be having a presentation followed by a discussion around the recent work of Alexi tulcinski a case for a Chaos Theory inclusion in neuropsychanalytic modeling so if you're watching live please feel free to add any questions into the live chat otherwise we will now have the presentation followed by a discussion with our excellent panelists so thank you everybody for joining and Alexi to you for the presentation um thank you uh Daniel um thank you Mark for finding the time I'm very grateful and excited to talk about this thank you Andrea uh Bryn Albert I hope maybe Mike will join us and uh thank you Daniel and active inference Institute for organizing this meeting so uh can you see the slides yep okay so I'll just uh proceed them um this is the outline of the talk I'll share some preliminary comments and talk briefly about assumptions in the current models summarize the data talk about free energy Principle formulations as compatible with chaos and talk about some future developments and hopefully we can discuss that I'm very much looking forward to have more questions than answers I'd like to start with what Mike 11 taught me I guess is just why it's important to me the most important thing is the clinical applications I think that with your work Mark and with yakpong sip's work we have been moving uh away from dualism over the course of some time but I think the other Big D determinist is still highly prevalent and dominant and I do think it influences power uh clinical work diagnosis and Psychotherapy and assessments and I think that it may be beneficial for some of the harder conditions we work with to consider including non-linear framework that's my main focus and this is the link to the preprint and the paper is published fully neuropsychanalysis Journal some of it is a discussion with this foundational paper that you mark wrote the new project for a scientific psychology General scheme which is a perhaps revision a Revival of Freud's famous paper project for scientific psychology and I'd like to give you a credit Mark that you have been opening conversations between disciplines that hadn't been talking to each other

prior to your efforts such as psychoanalysis and uh neurosciences you've been doing for more than 20 years with great success in the same spirit I wanted to mention that Society for chaos theory in psychology exists since 91 which is when the first conference happened uh about less than 20 years than the first conference on chaos in Lake Italy around Lake Italy Lake Cuomo in Italy in 76 which was mostly physicists at that time and then psychologists joined and I've attended the recent conference in Toronto which was absolutely wonderful and I met extraordinary clinicians and scientists and I think that it would be beneficial to open that dialogue as well and maybe if there is a wall between us to consider demolishing the wall um in terms of intent for the paper you mark pointed out to me that you've had an exchange with Robert Gallo to Levy for those of you who may not know him he's a psychoanalyst from Chicago a psychologic writer and a teacher who wrote a book on non-linear psychoanalysis and there's a deserves a lot of credit for his advocacy and you know he's sharing his thoughts he posed the question to you mark of possibly considering non-linear components in that seminal paper or project for scientific psychology and to which you responded that the key point in such discussion was whether the mental apparatus was indeed non-linear so I took your comment as a task as a challenger if you wish and decided to write a paper about it I don't know if the data presented is sufficient but I tried and very clearly I don't intend to contradict to criticize you know that would be funny of me to do I am hopeful maybe to add a branch of this tree that you're growing clinical and theoretical tree and should you consider it worthwhile to go in this direction we wouldn't have to change very much because this is the paper by Kristen Lancelot De Costa and others called stochastic chaos and Mark of blankets our financial principle is compatible with the uh chaotic attracting set what I thought was not present just yet was a formulation where the attracting set experiences a phase transition and I took Liberty to ask Carl about it and he shared some very nice thoughts that I wrote uh with his permission share at the end of The Talk I'd like to cite Maxwell ramstead whom you know and his colleagues from the paper and Basin mechanics of physics Often by beliefs where they talk about Dynamics mechanics and principles and in physics uh you know Kepler's work is descriptive it shows you know what the orbits of heavenly bodies are mechanics are the how of the Dynamics this is Newton's laws and principles are prescriptive they are about why things work this way there's a certain hierarchy in physics where you could derive the second law of Newton from principle of least action and then if you have initial conditions you could calculate the Dynamics the reason I'm mentioning that is I think Mark your new project for scientific psychology is a high road it seems to start from first principle such as the free energy

principle and you seem to derive things down from there so I thought why not try to see how the low road may look like if in addition to these derivations and deductions and Declaration of what entropy should be in the brain what if we measure it in animal studies and human studies in health and pathology and see what the data shows and whether or not the low road meets The High Road in the middle or perhaps something may need to be added I also like to clarify why I spend time in the paper talking about obvious things for everybody here the axioms and postulates and I'm quoting here Edward Frankel who is a mathematician at Berkeley and you know uh Michael will forgive me but you know a Viewpoint is that one can say that some parts of biology can be seen as based on chemistry which can be seen as based on physics which can be seen as based in mathematics so how do things work in mathematics we know that Carl Kristin likes to translate nearly everything he works on into the language of mathematics classically there's something called a formal system which we can metaphorically compare to building a house so the foundation of the house is a set of axioms that has certain qualities and then we build things up using the logic of Aristotle and we derive things like theorems an example would be euclidean Geometry which is on the plane that is based on five axioms of Euclid that's the foundation and then you could derive things like Pythagorean theorem now if we were to change just one of the axioms in the foundation the entire house changes an example would be the fifth Axiom of Euclid that parallel lines essentially don't intersect if you change that then we could have a house of on the sphere which is lobachevski rian geometry and meridians of the Earth are parallel and they intersect but you could have a parabolic shape where parallel lines also intersect so we'll have an entirely different framework and it may be naive to expect the theorem to look the same the figurative theorem if we you know put it on the sphere is different the formula is different um I'm mentioning that is because I I love multidisciplinary and I think that's the way of the future with that sometimes when we cross over into another discipline such as you know psychologists you know taking mathematical or physics ideas then we we may occasionally forget this formal system thing for example when we see things like human beings resist or violate the second law of thermodynamics then we uh the second law was formally related for closed systems in the state of equilibrium such as a sealed chamber of gas and human beings are neither closed nor in the state of equilibrium so it doesn't quite apply it's a different formal system one needs physics developing systems to make a statement like that so again in mathematics because you mark seem to have taken us deeply in the main of mathematics in your project you have differential equations for Action perception Precision you have you

know Mark of blankets random dynamical systems you know equivalences partial derivatives then in mathematics people fix the set of axioms they fix the definitions and then they build a theory right um so I want to be clear I want to clarify what are the axioms in the theory that is built what is the foundation and in the effort of time maybe I'll talk about just one of them uh which is Freud psychic determinism and this is a question for me not an answer I wanted to ask you because I am assuming uh whether you uh kept this Axiom or you changed it the Freud said in the Psychopathology of everyday life that there is nothing arbitrary or undetermined in the psychic life there's different versions of this statement and it has been repeated by generations of psychoanalysts over 100 years and it became very entrenched it's sort of a cultural Meme and a major tenet of psychoanalysis if I were to give an illustration perhaps those of us who are not in in psychologic therapy then let's imagine a analytic therapist hearing a patient who intends to say same with an S but they use the word shame right so such a phenomenon you know first of all I think psychodynamics psychologic therapist would probably say that's not an accident okay accident is is not naughty you know used term in psychoanalysis and secondly they will probably infer of what it might mean for example infer along the lines of unconscious or emotional Dynamics emotional conflict somebody else may infer that it may have neuropsychological reasons somebody else may infer that it probably is a migraine but what you might will not hear in psychodynamic culture is that that was you know an accident that it was due to chance I think the chance accidents Randomness stochasticity chaos are not used the bad form in psychoanalysis and that that is the illustration of I think for its assumption ultimately it's it's an axiom it's a belief that it's not questioned if we were to translate that assumption in the uh Kristen language this is the longevon equation from which Christian starts which is has notations by Daniel thank you Daniel and we'll have the change in the state of x equals the flow plus the Omega Omega is the stochastic component it's the uh you know in Daniel's metaphor we have a wave and then we have Ripple so we would have to change the stochastic component if we were to go with the Freud's Axiom and then if we do that we would find ourselves squarely in the domain of cold physics Newtonian physics LaPlace is demon LaPlace famously said give me a position and momentum of every particle and I'll tell you exactly where they have been before and exactly where they will be in the future now Freud didn't make statements like that but this is what would happen if you kill the stochastic component uh I want to be clear that it's not for its fault at all well first of all you need axioms to build a theory you chose that Axiom that's the theory he built and second of all it wasn't exactly mathematics and and finally the

Zeitgeist of the time was deterministic you know perhaps there were uh you know initial derivations of stochastic mechanics and quantum mechanics but for it when he was learning the world of deterministic physics was dominant and uh there's an important Nuance in this discussion pointed out to me by the first peer reviewer of the paper which said you know listen if you have something that changes like a stochastic thing and then you apply statistics to it such as variance then the variance of a changing variable will be deterministic and then if we look at things like verbs which are used a lot in talk therapy like Apple when Apple is a semantic memory it's a category it's a course grading already from green apples and red apples and all other kinds of apples so in that regard words are deterministic things so perhaps when we have a dream analysis session and the patient shares with an analyst you know the movie that they saw the messy you know irrational thing with images and tunes and anything and maybe words even if they narrate it then they they slow things down they average things out and they put it in deterministic tools which is words so with some objects we can apply deterministic tools but I want to point out that the foundation that everything and anything in the psychic life that all the limitations have no stochasticity Randomness is a very strong assumption and I'm curious what would happen if we were to allow ourselves to see things on this here on the parabolic shape it's not a bad exercise because the uh general theory of relativity would not have existed between without these other geometries uh in terms of definitions this is a familiar formula that you mark use of Shannon's entropy he originally wrote about Telegraph and Sandra and receiver in a fixed alphabet but here if we draw an attracting set which happens to be Lorenz attractor in Phase space and we look at these I don't know if you can see them they're small the white points you know that's the initial condition and then they spread out and evolve as time goes by but you can see the density of the attracting set it is a higher density and lower density so we could calculate uh probability distribution uh a slightly different question is this what would happen if you were to change the initial condition a little bit well one millimeter in Phase space different things can happen and one of them is shown in this diagram that would be Divergence in the formula here's initial condition multiplied by exponent of λT so when λ is positive you have diverges and it's called weapon of exponent and that's one of the criteria for chaos when it is negative we have convergence and when it is zero we have equidistant trajectories we have Newtonian mechanics this is an illustration of a chaotic system which is a Great Red Spot on Jupiter and it fits perfectly the for instance paper about the Markov blankets and stochastic chaos because that thing is stable we have been observing it for 300 years and one could say

that this elliptical shape is the mark of blanket around the chaos that maintains its integrity it's not a contradiction because in one spatial scale you have marker blanket and on another spatial scale you have a chaos within marker blank this is a convergent system that is a bit trivial to all of you this is a hole in the ground and then surrounded by heels in all sides and if you were to put a ball on the hill it's going to roll down in the potential well and typically in face space we'll call the bottom of the hole and a tractor and we call the heels repeller repeller is a term uh less frequently used but you know it's the highly unstable point in face space from which things move away from and they move toward the attraction now that's just one attraction the interesting things start happening if you've got two of them imagine a second hole immediately to the right and the boundary between them may very quickly become chaotic there's another illustration on a convergent system from the active inference textbook that I hadn't read when I was working on the paper I discovered it later the chart on the left is the density and again we see high density in the center and we have lower density in the periphery and the chart on the right is surprise we're very not surprised to find a ball in the center and we're very much surprised if it stays on the hill and I'll read you a citation active inference is restricted to systems like that one on the left which count to random fluctuations with their average flow and thereby retain their form over time it may sound like the statement contradicts what I said earlier that fep is compatible with chaos but it doesn't because you could have different spatial scales and convergence on one scale and Divergence in another I also historically think that we have been studying converging systems mostly so far now if we move from things like that which is very simple through a mind or Consciousness then I think that what which could be looking at is this that's a landscape with attractors and repellers that is very complex and I'm going to quote the former president of the Society for psychology is that that landscape is dynamic it moves perhaps some things on it are stable if or imagine a clinical scenario where there's a patient who is very prone to shame and very averse to express anger then we could say that shame is an attractor and anger is a repeller but and we work with pattern things in dynamic therapy so they could be relatively stable until the work in therapy makes them more shallow distributions but certain things change overnight some things change in a week or during the course of therapy so I mean strictly speaking that's a dynamic landscape and I don't think we'll model things this way um I'd like to also quote Maxwell ramstead from the same paper that free energy principle is a tale of two densities it would be incomplete to say that we just minimize one variable you know uh what happens is we minimize the Shannon's entropy of probability distribution over sensor States while at the

same time we maximize entropy of internal beliefs which is very important that's adaptation if somebody has rigid internal beliefs with sharp probabilities and then what happens they will have lower probability of survival they're not adaptable so adaptability means maximization of entropy of internal beliefs while maintaining the four means the minimization of Shannon's entropy of probability distribution over sensory States uh here's another term

comographic entropy that is not intuitive and it's from topology the picture you see is the photograph of the statue called intuition outside the Fields Institute of mathematics in Toronto here's a graphical illustration if you were to draw the trajectories in phase space and surrounded by certain volume say a circle with a radius ϵ and asked herself what happens to this volume as they evolve well when when it expands then you can have chaoticity when it uh becomes more compact you could have convergence when it stays the same you have periodic system which is indicated on the chart on the right where you have you know a correlative they call logarithm entropy on the y-axis and you have this ϵ on the x-axis so the flat line at the bottom where the the circle doesn't change is the periodic system

homographs an entropy stays zero the curved line in the center that converges on a k is the proper chaos this is theoretical chaos and the options from it uh is is what Kristen calls stochastic chaos this is chaos plus noise and then you can have pure nodes and then you can have pure linearity so all of these things exist they're all on the table you can have geometry on the plane geometry in the sphere geometry in the parabolic shape so what is the data show and uh when we look at EG entropy people use approximate entropy permutation entropy and spectrometrically which are all derivations of Komogorovsinai and the first two are in a different phase space from the last one I believe Mark correct me if I'm wrong your side go series uh a study in your book Hidden Spring when you talk about entropy and phases of sleep and they use Spectrum entropy which is a derivation of chromogorovsinai in the frequency domain analysis phase space so the data shows that you know uh systems for as small as an axon of a squid isolated in the lab to single neurons two coupled neurons to EEG we have evidence of positive lag enough exponents in some regimes of operation with EEG we can be more specific that Alpha Rhythm seems to be weakly non-linear while things below Alpha like Delta seem to be more periodic with noise and things above Alpha such as gamma and high gamma seem to be chaotic so if we were to summarize this data that it would be we can say that komogorovsinai entropy appears to increase as generalized arousal increases from Delta to gamma rhythm seems to be higher in healthy alert brain mind functioning than in the states of coma seizure deep sleep and it can stay close to zero in a coma or deep sleep the interesting

anecdote here is that a simple Act of closing your eyes when you're alert decreases komogorous in ianentropy clinical applications are vast and developing fast for example but except for Clinical Psychology I must say and Psychiatry and psychotherapy but in neurology the seizure detection on the EEG is right now can be done with AI with about the same accuracy as a human neurologist and that algorithm does use claoticity assessment there's models for Parkinson's neuropathic pain and other have other examples that I'll skip right now this is a graphical illustration of phases of sleep and again it has called mogorithm entropy as a phase of sleep it shows wakefulness and then shallow sleep progressively deeper sleep and REM and that's the time Evolution if you move them in frequency to where it belongs to wakefulness then you'll see the same progression that komogorous and entropy appears to increase as a function of generalized arousal this is a seizure onset at about second floor teacher is a reduction in chaoticity healthy functioning is higher characteristic than a seizure this is a coma which is relatively invariant fat in Delta frequency this is a nearly periodic system with some noise and death is a linear system it doesn't change this is persistent vegetative state minimally conscious state where researchers were asking a question on prognosis of survival and they've concluded that diversity and variability of the EEG was the predictor of survival higher entropy means life so the hypothesis is that chaotic stochastic and linear processes as well as hybrid ones such as primarily keratic functioning with noise can be present concurrently at different scales of the brain mind or the same scale but in different places or at different times I realize it's a mouthful but it fits the data presented in the paper this is again a picture from the textbook and after inference by Thomas Parr Carl friston and Giovanni pazulu it's a cortical column and the errors you see here uh ascending and descending connections if by chance you're not familiar with the framework they correspond to uh the ascending errors are prediction errors and descending ones are predictions this is what they say in the book that there's an asymmetry in message passing and the reason for that is that the operations required to compute prediction error from expectation are non-linear this non-linearity is due to computation of predictions using non-linear functions that tend to increase the frequency of the signal I'd like to talk a little bit about reduction it became very fashionable and sometimes silly to you know use this like oh your reduction is now your reduction is done the second most popular term is oversimplify you'll oversimplify now you oversimplifying to me they're not bad words and they're necessary tools to simplify and to reduce uh you know the map that equals the territory exactly is useless nobody's going to use it it must kind of I'm quoting James Glick simplify and abstract but you know I'll illustrate the

the specific term if the car is moving with a fixed velocity in a straight line and if I know exactly where it is right now I can tell with certainty where it has been an hour before that is a reduction in time and I can do that because the formula is linear its distance equals velocity multiplied by time and velocity is constant if the watch is broken and I disassemble it and I find a faulty part and then I put it back together uh and it works then it's a reduction in space and I can do that because watch is not an ant colony uh to use Daniel's terms which is uh equal to the collection of its parts so I wanted to talk about Mark your uh mention of the predictive hierarchy composed of billions of homeostats in your in your work and this is approximate picture this is not the exact picture of what you talk about in a new project and this is from your Hidden Spring it's a predictive hierarchy where predictions flow from left to right and prediction errors from right to left and uh one of the statements in the book in The Hidden Spring you say that brains many complex functions really can ultimately be reduced to a few simple mechanisms like this so it appears that you you know you call it reduction but I just wanted it is a powerful model it is a flexible model I wanted to describe some features of it well first of all if homeostat is present at every single scale then we have this homogeneity of mechanisms throughout the scales and I want to say that in physics things don't look this way quantum mechanics is the microphysics statistical mechanics is a molecular level Newtonian mechanics is heavenly bodies and then we have general theory of relativity there's qualitative shifts there isn't a homogeneity of mechanism all the way throughout in fact when people say take entanglement and they apply to human beings that's sort of a misunderstanding or entanglement uh the second comment is from Michael Levin who talks about these scales and if you know his comment is that you could have different functional spaces at every scale where you could solve the genetic task on one scale electrochemical in another morphological and a third one and then how you do message passing what is the common language it's not a trivial question along the same lines I've already mentioned that a message between the scales can be non-linear if you do prediction errors now if you take one non-linear message and then you multiply it by billions you're going to have a full on turbulence which is a chaotic system it's a dynamical system and turbulence you cannot reduce the homeostat or anything else you know saying that water consists of H₂O does not explain turbulence so I don't know that's an open question but I wanted to maybe share some perceptions of this model this I talked about uh and this is the for instance thought about phase transition and the attracting set he said that if you have a hierarchy of lorenza tractors a slower one and a faster one and the slower one governing the transitions of the faster lorenza tractor and then you could have a pep

formulation with an attracting set experiencing changes in transitions and that's brilliant I think and it matches the anatomy because the brain stem and the subcortical structures appear to be slower lorenza tractor that could be seen as governing the transitions of the cortex which is astrological tractor I know I'm oversupplying things a great deal but it sort of this is his thoughts I think him and Lancelot decoster working on that perhaps Maxwell ramstad's group is working on that as well the clinical applications the landscape that I've mentioned before would be absolutely lovely to see if it were to consider going there uh borderline personality disorder or another term for that could be complex trauma the metaphor that we use for these patients is they're predictably unpredictable that the only thing we know on Wednesday that we don't know what's going to come up suggesting characteristicity acute trauma has a symptom of hyperarousal suggesting possibly higher level of chaoticity an onset of mania can be seen as a transition into a higher energy State and addiction is particularly with stimulants can be seen as phase transitions into a chaotic state it may sound like what I said now is contradictory to the what I said before when I suggested that you know lower characteristicity could be a problem such as an onset of a seizure and here I'm saying higher chaoticity can be a problem but that would only be a contradiction in the linear world in the real world where linear is very small percentage of what's happening I'm saying both that too little and too much chaos could be a problem just enough chaos imbalance could be perfectly fine this is a graphical illustration of what can be done and this is from my colleague and friend David Pinkus who is a professor in Chapman University and I think also former president of the society for chaos and psychology Society these are trees and may sound like completely unrelated to what we do except if you look at a human lung or a dendrites in a neuron and you can see a healthy Tree on the right from me and the nearly dead tree on the left and if you do the fractal assessment fractal Dimension assessment on them by counting the thickness of branches and count the number of branches at each level then you can quantify and you then you can have approximate quantification of health and resilience versus xenopathology and again as exotic as it may sound this is very similar to what people do when they detect a seizure on the AG they measure CD correlational Dimension which is approximately fractal dimension and I think my 30 minutes are up thank you so much for your attention thank you Lexi gray presentation well looking forward to this discussion first Mark please feel free to add your Reflections then Andrea then we'll continue on thanks very much I when Alexi asked me if I would join this panel uh the first thing that I said to him was yes uh coupled with the second thing being but please be aware that I know next to nothing about Chaos Theory so

you know I'm not sure how much uh of a contribution I can make to a technical discussion um of this kind uh so I'm just repeating that uh before I say anything else if I say anything that sounds amateur that it sounds as if this guy doesn't know what he's talking about when it comes to Chaos Theory it's because I'm an amateur and I don't know what I'm talking about when it comes to chaos theory that said um the second thing I want to say is I'm not entirely sure what is at stake uh here you know it seems as if Alexi feels that it's really important that um that I and uh our colleagues uh get our head around chaos theory and bring um some of its um what it has to offer uh into our field um because I don't know much about it I don't know what's at stake you see I don't know uh why that's important so uh um you know perhaps somewhere along the line in the next uh just less than an hour that we have perhaps Alexi you can clarify that for us um the third thing I wanted to say is um and perhaps in order to illustrate the first thing is that as Alexi was talking now it seemed as if he was talking about chaos as if it were synonymous with stochasticity or or Randomness and that's not my understanding my very limited understanding of chaos theory was that it's not a synonymous chaos as used in that um in that science or in that theory um is not synonymous with with Randomness it's rather synonymous with unpredictability um because we don't know the initial conditions of the system what we can't specify all the variables with sufficient Precision uh in their initial conditions it becomes impossible to predict um and and because of the nature of the interrelated relationships of the variables one tiny little error in your estimation of the initial conditions can lead to massive consequences and this is why it's so hard to predict uh chaotic systems so if that is what is meant then I would say um of course this applies to the way that the mind works and the way that the brain works um but that leads me back to my second point I'm not sure what's at stake but I would say if that's what you're asking if that's what you're claiming rather uh Alexi then I would say that's that surely must be so that it is exceedingly difficult to predict um mental and neural events at any level of complexity um now uh I think I've said three things let me now say a fourth thing um I think that a crucial um well let me put it this way uh of course what the brain and mind are trying to do it's a matter of trying to predict what's going to happen so when I what I've just said about how how difficult it is uh to predict what's going to happen in the brain or mind it's it's it's it's almost the same thing as to say uh that the the mind and brain are trying to model the world uh that's that's the main thing that we're trying to do is is is is uh have a good model of the world so that we can get some sort of grip uh on this chaos um so that we can act rationally in the sense that we can act um in order to preserve ourselves that we do not dissipate uh by uh just uh exploring all

possible States and no longer existing as a system because we as systems need to remain within certain bounds across a great many variables as you said um we can't afford to just let should happen and you know and throw ourselves to its uh to to to its Fates uh we need to try to predict uh in a very unpredictable world uh what must I do in order to meet this need given how the world works so the brain is trying to model the world uh our predictive model uh is is trying to synchronize itself with the world and for that reason it's going to be very chaotic um but the crucial point is it's trying to reduce uh the unpredictability um with that said uh I'll make a fifth point or maybe it's another part of my fourth point no I think it's a fifth point which is that the way that I conceptualize the predictive model and it's not me alone who does this is that it is a hierarchical model um and I think that's very important and perhaps here's where things do start where it starts to um at least in my mind I start to see some of what's at stake um I I think that the more peripheral layers of the predictive model I see the predictive model as a as a not as a triangle but rather as an onion so the core is the the deepest layers of the model that is to say um the the layers that have the um the greatest um spatial and temporal generalizability um and they're the most stable layers of the model and as we move toward the periphery which coincides with the periphery um of the of the nervous system in other words the encounter with the outside world so it becomes more chaotic we are able to talk we can tolerate um more unpredictability at the periphery uh and uh we we're trying to and that's and that also allows for greater accuracy because the world is so unpredictable we have a more more chaotic uh um outer layer of our predictive model where where things also there's less that's at stake you know is that car going to turn left or right well I thought it was going to turn left turn right so what doesn't matter you know but um it is the law going to is the car going to obey the laws of gravity that that matters more you know that's not something uh that that's a much more consequential surprise uh if it if it does not um and perhaps that's a silly uh analogy but at least it demonstrates the general Point our most heartfelt beliefs um and um uh that is to say the deepest layers of our of our predictive hierarchy there we find the same patterns over and again and we we we we find our soul we we are more certain um of our predictions and we act more stereotypically in accordance with them uh whereas uh at the periphery uh that does not apply so I think um at one point in your presentation Alexi you you were saying something like you know one size doesn't fit all it's not all just one mechanism all the way down uh of course I completely agree with that um the the the it's not just one mechanism and the different homeostats that you were referring to on their different scales are also serving different needs you know and there's there's not a common currency uh I I think

that's a very important part of um of how Consciousness works why we have qualia at all is because we can't introduce things to a common currency we're dealing with categorical variables uh at the very least in relation to our bodily needs they are categorically distinct needs and they need to be treated as such and this is why we have different feelings in relation to these different uh categories of free energy so um those are those are some of the main things um I I wanted to say just to get the ball rolling I'm mindful of the fact that Andrea Clarichi uh only has uh 15 more minutes with us so um maybe we can pick I can pick up on some other points later uh but at this stage let me hand over to Andrea thank you thank you can you hear me now yes sorry I will try also to articulate something of the very fascinating paper of Alexia I must confess since I'm also a psychoanalytic psychotherapist a psychiatrist it's not my field I'm not a physicist so I can really say a few words some is of as I said already Fascination because in general I hate to stick to rigid theories so if there is a paper like Alexia who can caotimize my theories even in psychoanalysis in the workings of the mind and the Brain I'm happy to to to to to dive into it at the same time uh I unders I I don't have the skills to to to to discuss the the math part of his papers or even the the physical formulas and so on but I was reminded of a very early and you I think you all know that book uh Jacques Monod Chants a Necessity which is a classic and I think it's a book in which you uh you can from from scratch I mean here that we our aim in the world and I I I'm I know I'm talk I'm sounding uh teleological but we have teleological uh um teleological uh AIMS in order to stay alive we have to face chance and cows so I think this is uh in order to survive and reproduce we have to uh have a mathematical system to reduce chaos and to bring up again the determinism or psychic determinism since one of the of the points that Lexier I think raised is do can this can we still hold the point of Freud's about psychic determinism and I think if you apply not to the physics of of just materialism of of of things actually but to the living organism I think we do this is my point and I I would like to uh to hear what Alexia says in the end the chaos theory mathematics as far as I understand it must be a mathematics who reduce chance or chaos to uh to a linearity at least in the in the living organism and I and I wonder if and what Alexei thinks about it because you seems to disagree with that want to give a direct response Lexi or ten brain and Albert be awesome to hear your takes as well some of the thoughts from Mark and from Andrea today she started by responding to Andreas women's remarks or yeah maybe I'll start by because I think the key point that Mark you brought up is what's at stake is still unclear my impression isn't please correct me if I'm wrong I was very influenced marked by your paper I forgot exactly the journal where you talked about Psychiatry world how these sort

of you know convention-based diagnosis appears and major depressive disorder here's a bipolar disorder nothing of which has to do with etiology you know uh convention and statistics you know and then you brought up this you know revolutionary thing with effective Neuroscience where you talked about the manic State and the depression State and you know tied into the punk sep system I think is extraordinary but what I wanted to say is you know if we look at how we diagnose and think about what's happening and we look at the language of how we do it it is still a deterministic language at the end of the day so when we moved from dsm-4 to DSM-5 we brilliantly discovered that continuous variables exist we used to have buckets everything was in a separate bucket and then we said autism as a spectrum and everybody applaud it but you know mathematicians were quietly laughing in the corner you know because of course continuous variables exist and but we don't do Dynamics we're still very static here's this clear-cut thing and this is the clear-cut continuous thing right and I'm my impression I may be wrong is that even in in our neuropsychanalytic developments we still do things deterministically we assess which bucket things belong to maybe two buckets at the same time or three buckets but we don't do things like you know how the chaotic systems are qualitatively different from aesthetic things I'm quoting William Solis they have transients they have contextuality and they have emergence we don't do that you know and uh I think that if we embrace the chaos theory if we do things like correlational Dimension assessment or fractal Dimension assessment starting from this system of classification of emotions there's still buckets here's a clear-cut thing called fear and that's a clear-cut thing called rage buckets determinism okay and you know they look differently in the dynamical systems world and then the system of Psychopathology could look like you know correlational Dimension changed and not that somebody fell into a bucket you know if we look at this complex you know landscape of attractors and repellers and examine psychodynamic things like please forgive me if I'm incorrect uh um but one of the possible definitions of repression could be seen as not letting things into Consciousness maybe we can call it suppression but what I'm trying to say is I don't want to think that right now it hurts it's a painful thing that in in the terminology that I shared is a repeller and perhaps when you work through things in a psychodynamic psychotherapy perhaps you put a scar tissue on the wound it becomes less of a Sharp you know uh repeller and it Smooths out you know we don't talk we don't diagnose like that we don't trip like that we don't think like that uh one paper that I'm working on that is in in peer review right now and I can't share too much says that you know we think like storytellers I'm quoting Daniel Kahneman our our you know thinking Mind of a clinician is not a

statistician we don't think in probabilities we think in labels and we think in stories and I know you have this powerful quote mark from Freud who said that that's the nature of the mind and I most respectfully disagree that that's the entire nature of the mind I do think that we have things on the mind that are not stories and not words so if the nature of the phenomena is chaotic and stochastic and periodic then we can use appropriate mechanisms to diagnose and classify that fits the the nature of the model the second thing you said case is stochasticity I absolutely agree with you that they're different you know the chart that I've shown if you converge to a stable uh stochastic chaos is deterministic for a deterministic so if initial conditions are changing they will diverge but you can predict things forward not back but uh there is no formula in a stochastic thing you know you throw the coin you don't know you have no predictability whatsoever short term or third or long term so they are different and in the sense that you're talking to sensitivity to initial conditions this is music to my heart to hear you to say that you acknowledge that irriticity clearly is present of the brain on which level of the hierarchy in your model is a very important thing but again when we draw this picture whether it's a triangle or an onion with a core and layers it's a static model and when you acknowledge that there's higher unpredictability toward the periphery it could be but there could be Dynamics about the structure of the model and perhaps what you're saying is survival you know like I must breathe and I must breathe air and I'm not in the water and that is a very stable prediction absolutely but when we deal with Psychopathology and we deal with things like shame and pride and awe and other things then I wonder if we should continue to use this metaphor of homeostat I really don't see how homeostat applies to gamma rhythm in the brain or to shame so and homeostat is is the convergent systems so that you you minimize entropy by keeping things in the range however if you dive inside the homeostat if you look at the temperature variation inside our narrow range you may see chaos which is exactly the picture that I saw so um that I showed um hmm and to Andrea's point is about the Forex principle of psychic determinism yes I almost is thinking that we haven't said goodbye to that we haven't tried to say that chance exists in the mental life accidents exist we don't do that there's a firm tradition in and psychoanalytic Psychotherapy to say there's no accidents it's all determined that's a major key point of the discussion I don't know if I said too much maybe Andre if you want to give any yes or how do you how do you want things to continue in the coming minutes and years okay thank you thank you Daniel uh just adding a word that I didn't use because I think it's very important about trying to defend the point about psychic determinism which I'm I'm very fond of I mean the word is adaptation I have difficulties in because when

you see a psychotic patient when you see a borderline patient well I don't see chaos or total unpredictability idea the adaptation their disruptions uh I I try to to to to to understand what are the even the mathematics the mathematical part be behind that in my opinion has to have a tendency toward an order which is adaptive in this way I see uh the the mathematical part of of chaos may be of use I mean but bringing uh to an explanation not what maybe not to an order or to a determinism but to a point in which uh there must be this adaptive reason even in the chaos do you see my point I do but I think there's a deeper issue about how we think you know say there's a clinician psychologic psychotherapist we have a work in memory and The Miller's law that Mark you quote a lot about the seven units of information uh in the auditory verbal working memory I think in The Miller's paper he actually said seven chunks so I'm I'm sharing something from the paper in review right now we cannot think probabilistically because it overloads the working memory this is why when I'm a clinician I think about psychosis I come up with a story a formulation that fits in the work in memory and this influences our work we don't have the mathematics in here to think in the terms of probabilistic things and Chaos Theory and other things we need tools to do that and in mainstream neurology they use tools they do talk to the patient that examine the patient but they have MRI they have ehe they have pestican and all the other things in psychologic Psychotherapy will say we don't need any tools we just talk to a person that is sufficient for everything we do but that's not necessarily true Beatrice BB who is a psychoanalyst works with Mother infant diet and there's zero words exchanged zilch but she looks at this movie that she records on a video camera and then she throws it shows it in a slow motion so she takes the flow which is dynamic flow not words not discrete things and then she slows it down and she shows the level of Attunement between the mother and the child we don't do things like that we talk and we use this deterministic things to understand the Psychopathology of the patients and not psychosis and that's schizophrenia and that is bipolar one this is panic system and this is seeking system we determinize everything and that is a limitation because inside I believe what is the fear Cascade how does Fear start it's a flow it's a turbulent thing and we put one label and it say fear we've classified it so for some things it's appropriate but for some things that may not be it may be you know as as you know Einstein said as Mark you quoted Einstein as simple as possible but not simpler than that so at which point we are missing with or missing the phenomenon by using deterministic Tools in our work thank you thank you Daniel and Mark to everyone thank you I have to go awesome so Albert and then Brennan will continue on this topic yeah thank you for the presentation and a very interesting discussion so

far um one thing that comes to mind is the the the well the thing we're talking about right now the psychic determinism as as uh depicted as being a foundational law and psychoanalysis well I'm thinking about the authors the psychoanalytic authors who came The prominent ones who came after Freud uh who seem to have a uh who have room or this this chaos maybe they didn't use this term explicitly but I'm thinking for example Wilfred Beyond who's got the notion of the O layer um which has connotations of uh well the thing that cannot be known or categorized who has like in indeterminate elements of the mind and that's one of his major contributions contributions to psychoanalysis on the other hand the more the French schools I'm thinking of Jack Lacon who's got the notion of the real which is also one of his major psychoanalysis where Notions of rupture of absence or of lack are very prominent I'm not sure I'm not an expert here but I think La carre explicitly uh challenges the notion of Freud's mechanistic worldview in one of its seminars so I'm wondering what you think about this Alexei hey I'm not an expert in Lacon at all and I know Wilfred Beyond in a limited fashion so perhaps Mark uh is a better person to to to to uh to talk and I I like for it particularly it marks interpretations and translations but maybe Mark can comment on lacana and Beyond I I'm not sufficiently knowledgeable about their work if you want to mark or Brandon you can share some thoughts yeah please um well I think that uh um the the point that Elder is making that is we must remember that psychoanalysis is not synonymous with Freud um that not everything in psychoanalysis stands or Falls by whether or not we accept this or that tenant to Freud's um and uh in neuropsychoanalysis we try very much to draw upon all of the Traditions uh within psychoanalysis and we we sort of uh fondly believe that we'll be able to find more a common ground because there are greater empirical tools available in neuropsychoanalysis than in psychoanalysis by itself you know we're we're hoping to resolve some of these uh theoretical disagreements trying to move Beyond just different systems of belief trying to move more toward a a psychoanalysis in which we can decide questions in an ordinary scientific way but it will always be very difficult for exactly the reason that Alexis whole talk is about I suppose which is you know the the extreme complexity uh and therefore um unpredictability uh of the of the phenomena that we're dealing with um so I think Albert's points are valid uh that uh the Beyond's approach lacon's approach um they they had they add a great deal that they they don't um they don't um have as much to to lose if we throw out the principle of psychical um but having said that let me just quickly say a word about psychical determinism because this has been a a big part of what uh Andrea picked up on on in what Alexi said um I I think it comes down to the the the question that I asked at the outset it's not a matter of is the behavior

of the Mind random uh or uh it's it's a it's a question of how predictable uh it's not a matter of is there no determinism it's a matter of how easily can we Trace effects back to their causes um so the fact that there's a great deal that goes on where we can't do that uh there are many things our patients say whether they be slips of the tongue or anything else of that kind and not only to do with parapraxes but anything that they do you know there's certain things that we see with regularity I mean the very phenomenon of transference is all about regularities that there's certain things that we see Yep this size this output this I I can understand um this happens repeatedly uh I I I I think I can explain that with reference to uh certain aspects of the patients in a world and and their past but there's a great deal else that we can't so I don't think that it's a kind of you know either there's determinism or there's or there's a stochasticity I think that there's this is why I was talking also about the layers of the predictive hierarchy the the deeper layers are the unconscious ones their things are pretty pretty stereotyped um pretty pretty predictable in fact too predictable I mean in other words it's over generalizes it it's it's it's too confident about its beliefs um but as we head toward uh the pre-conscious and conscious peripheries you know their things are much less predictable um so that's that's I I won't say more about this a couple of things in what Alexi said in his uh in his comments after mine that that if we have time I'll come back to you but I I think let's rather hear from from Bryn before uh I say anything more yeah thank you thank you so much um I I I'm perhaps the least qualified here to to comment on your on your paper uh Alexi um but I do have one or two thoughts and they they come from a place of being very fortunate uh that's why my life to have been a patient from for many many hours um sometimes on the couch and also to be a clinical psychologist I failed to mention that at the at the in my introduction so I'm I'm a clinician I was a until recently uh when my foray began into more um research based areas and via the active uh inference Institute and Mark's Work and uh and and and and called Kristen's work I was a a pure clinician I had it in my mind that I was going to treat Psychopathology to the best of my ability and but really it was being a patience first and foremost that brought me into closer contact with the the theme of your paper and certainly the themes of Mark's Work um and I think I've encountered it from both areas and it certainly it matters to or to my mind it matters less the inner workings of the degree to which uh chaos stochastic uh all these big wonderful terms that that aren't only beginning to to even think about but um oh what what's what matters to me is that how do you help somebody uh who is in a in a state who comes in disheveled and and and you know how does one go about helping uh aiding somebody in that state um I have um fell out of love with psychoanalysis uh for for

personal reasons um but it was in returning to it I I've become more kind of a better acquainted with its strengths and its limitations and I'm realizing that it's um it's that when you help somebody and manage to reduce the noise to reduce uh the chaos in somebody's mind um that they that when they become the greater degrees of Health start returning to them that they they do rely far more whether they know it or not on probabilistic reasoning in their daily um kind of functioning um and so I think there is a strong case to be made for a mathematical uh way of processing the world it doesn't have to necessarily be conscious made conscious to the person um it can be a bit overwhelming to uh to try and impart that on somebody but um I think a brain that is not working properly uh regardless of the degree to which there is chaos that can be measured or quantifiably uh measured in that brain somebody isn't is tripping over themselves because they um they've they are unable to predict what might happen and and that is a terrifying Prospect because I think there is a some Reliance and some some staunch dependability on the the fact that the sun is going to rise and that it's going to set and that I can then focus on other things uh um so that's just the my comments more from kind of a personal front of being a a patient uh and and encountering the limitations of psychotherapy and what what I would have what I would have done in those moments of need to have not as you say had a more a deterministic view of what was happening and I've tried to um transfer that or carry that over rightly or wrongly um into my own therapy as a psychotherapist as um so a more personal thing of what it means to be to feel chaotic and what I see regardless of how confirmed it is um you know or not um but but chaos Reigns and and it's to try and not compound that uh it's certainly something that um I'm interested in so yeah thank you Bryn I'll also give my thought on what's at stake and what's on the table as a total non-clination but what I saw in Alexis paper and also what I see people coming towards I see the paper as Alexi making a a real stable of concept or portfolio of topics ranging from the first principles of active inference like surprise bounding to Dynamic stochastic chaotic which is not the same thing but they sometimes cohabitate so bringing a set of Concepts into play making it matter for the field so that it can matter for the clinician so that it can matter for the patient which is where you totally brought it home to Bryn and in my totally simplified model I was seeing all this diagnostic Machinery from personalized data EEG recordings all these surveys and samples all the different conversation therapy itself all these different measurements inference and then it passes through the Markov blanket of the diagnosis and it's like well now we know that that person is X or they have X and then here's where we now Branch back out with the modalities of of treatment and so that bottlenecking has obvious relevance in terms of like

getting a categorical grip on what we basically all agree are wet wear complex systems they're sensitive to initial conditions all of that what if though like you brought up the example with a scar tissue maybe it's literal bodily Scar Tissue maybe it's conceptual Scar Tissue what if that inference were able to be understood in the context of an attractor landscape and it's not just a singular attractor so we have multiple tractors we have a chaotic landscape and so what if we were able to um perforate that blanket speaking super Loosely to bring in what we have already recognized as diagnostically relevant like the fact that patients who have differential entropy have this or that epileptic outcome or a coma outcome bring in that Machinery that led to the diagnosis but now take it beyond the diagnosis and maybe even beyond the clinicians concept directly into the hands of the patient thank you so much perhaps we can hear from Mark and uh yeah I am not eager to talk I'll if there is time I'll say something but if not that's okay um well let me first of all say something about my um screen you might some of you have noticed that suddenly everything went black behind me that's because I had I'm in South Africa where they have power problems and then um a kind person in my household thought that they would help and they've put this huge Spotlight here which is shining in my eyes and uh um so um if I if I look as if I'm in a rabbit in headlights it's because I am I sent a text message saying please come and remove this big lamp you put here but they don't seem to have received the message um now back to our our actual uh topic um to the extent that I'm able to not be distracted from it I think that um that when I asked um uh Alexi what was at stake um his main answer uh was to do with a diagnosis now I don't mean that he's saying that that's all that's at stake but I think he's saying this illustrates what's at stake um so I want to first of all make sure that because I'm not sure who our audience is and who's going to be watching this in case anybody uh is misunderstands uh I want to make clear that in psychoanalysis we've long ago moved away from DSM and ICD type diagnosis uh we we we always thought that um the the the nosology uh the the psychiatric nosology was was who really artificial and created divisions where they don't exist in nature and we replaced it with a far more Dynamic approach to diagnosis and it's not just a matter of labeling it's a matter also of what kind of thought process lies behind the diagnosis in other words conceptualizing what's going on here rather than saying this is a an instance of personality disorder and I'll specify which type you know we don't think like that in psychoanalysis and and really we never have um in neuropsychanalysis uh the the we we build in that same tradition um I think that what Alexia was uh was worrying about is he's he's saying yeah but you know you use you use systems uh you you say that there's this very there's a certain number of drives um and

you use those to pass what you seeing clinically and yes we do um and we have to because you can't just say what I'm seeing is chaos you know what you've got to say is well what I'm seeing is chaos but let me try my best to find the deeper structure of this uh and so what we do is much the same as what I was saying earlier about how brain and mind work likewise the clinician's mind works that way we have some deeper Concepts uh that organize what we see and then more on the surface uh we we we see uh the the complexity you know in other words we don't think that we can reduce a patient just to one saying this patient has problems with this drive or something it's not as simple as that of course but we do start with saying well these are the drives these are the attractors uh to put it uh in terms of the attractor landscape that you're talking about these are the these are the preferred States um of the of the human phenotype you know and the and in mental life too they are preferred States there's certain there's certain attractors uh that that we're trying to stay in those in those um like for example uh if fear we want to be safe we don't want to feel fear uh you know with respect to attachment bonding we want to stay close to our uh to our attachment figures we don't want to lose them with respect to rage we we don't want to be frustrated and have things standing in our way we want to we want to be able to have free access to the objects of our needs and so on you know those are these attractors and you know we we have great difficulty staying in those in those homeostatic bounds and uh so you know it's a it is a dynamic a picture and then it's a quick over over the question of the of the different drives there's the the highly individualized predictive model that each one of us comes up with in relay relation to those drives and in relation to the conflicts between those drives and so on so I think that um that this way of thinking that we use in psychoanalysis and neuropsychanalysis is already of the kind that Alex is calling for so I think that what Daniels just said is that therefore perhaps uh very important to make clear uh it's that that he's offering us a language an additional language an additional I don't just mean language I should say an additional set of Concepts uh and um and a highly developed one uh which we might find useful and so we should take this on board that sounds very convincing to me that if if that's what's at stake here that that Alex is saying this is a language that lends itself to the phenomena that you study and to the way in which you go about studying it then it sounds to me as if Alex is right um so that's the main thing I wanted to say in response to to his comments uh uh in in relation to what I said um if anyone else wants to say anything else that's that's good with me but Alexi um if if I've misunderstood and if there's something else that you think is at stake that I'm not getting now's your chance to to to say it well I I just wanted to to end the conversation when you said that that if this

language might be of any even to consider this language to to in our future models I'm a happy man and there's nothing else I need to say uh um I just wanted to to say that these are human things they're occupational hazards some of the things I'm talking about this paper I'm working on about narrative stuff is that you know uh of course you know it is much easier for us to think in in uh categories now even the psychoanalytic world you know what what uh you know we do in in the old let's take the mcquilliam system and the previous one we used to say three levels of functioning you know uh neurotic borderline psychotic these are categories they're not DSM categories they're not bipolar one but they're categories nonetheless and then we have character Styles obsessive character style histrionic character style their categories okay so to some degree it's a human condition to put things in categories and I do think that this is this is a very strong historical reasons that we think this way we we this we're we have generations of thinking this way and categories and yes at the end of the day when we paint a portrait of a patient it's developed and it's like a tapestry all of these things weave into one another and it's complex but we start from basic categories and you know um certainly if we consider that you know some things can be transient there can be emergence they're like that the certain behavior in one context is an uh compulsion but it's not a compulsion in another context and perhaps I'm saying trivial things to clinician and to clinicians and uh also what what dream you said just a comment when you said as a patient to reduce the noise and reduce the chaos I detect the bias in the language you see how we call you know Psychopathology and DSM disorders disorder is a bad word for us and I'm trying to say chaos is healthy I spend most of my day in gamma functioning which is a chaotic State it's not a bad thing so in the folk language chaos is a bad word disorder is a bit I want to put things in order and then perhaps we're missing something what comes to mind is I think Mark you had meetings with Michael Levin and Ian mcgilchrist who is you know big on talking about the mechanistic you know logical linear straight and narrow left hemisphere type of course it's not left hemisphere but that's the model versus holistic sensorial kind of right hemisphere artistic type and we're missing the second part like Daniel's background is surrounded by Art where you know we we spend less time in the right hemisphere maybe maybe appropriately so but I think that there's some things that lend themselves to that level of understanding and we would break things we we we look at Psychopathology and human suffering like it's a carburetor sometimes where we again you know put it in buckets that's my impression but what you said uh is is wonderful and I I you know I don't contradict with anything I agree with you said Mark so uh Alex if I can just um comment uh if you know if I if it's almost like an arc you know the the pot of gold

at the end of the arc is is the uh reduced chaos the reduced noise in terms of the patience from the patient's perspective but how you get there ironically is actually by inducing um uh States of uh tolerable chaos um and then kind of re-reformulating that reconsolidating that chaos um so it's not that uh that from the you know healing or growth or whatever the patient might present to you for it has everything to do with um minimal or marginal chaos induction and then you know reintegration of that so it the whole process is is premised on chaos but in bite-sized doses maybe one could say right and just to finish up and then I'll stop talking Jonathan schadler who is you all know and he's uh you know rightly uh respected as a psychoanalytic practitioner and teacher and writer uh he recently said that take a patient ax with an analyst why together they are emergent properties that is not the same thing as patient x with the analyst Z right and show me one study where we take that into account we don't we just we just do random you know a clinical trial then we do meta-analysis of random clinical trial what technique did the therapist use they used you know whatever four-letter therapy they applied nobody looks at the emerging properties in a dyad the this Matrix you know between the two and that's the language of dynamical systems this is Daniel Friedman you know ant colony you know a single ant doesn't build Bridges and grow mushrooms and Colony does all those things we don't talk about emergence in psychoanalytic Psychotherapy do we it's true it's also a fundamental challenge after all you can't replicate within one dyad or across dyads in a finite world so so it does point to fundamentals in our last minutes if um each would like to give some thoughts where do we go as we now the open of exponent away from each other in our paths so maybe um Albert Brin Mark and then Alexi what are your overall thoughts or Reflections and then where are we heading go for it Albert um yeah I was I was about to comment on Alexey um about the emerging properties thing um it's it's it's kind of the same point I made earlier in in later psychologic theories do talk about emerging properties um but yes I think as what what Professor sulbs just mentioned uh this this language I mean this this new set of Concepts makes it very uh practical for us to communicate with each other because psychoanalysis has a bad habit of Reinventing new Concepts every every new decade so it's there's a lot of chaos to use that word um within uh psychonalytic discourse so yeah but overall I thought I'm very grateful I think this was a very interesting talk so yeah thank you thank you yes so uh by way of closing uh remarks I I I agree with um actually it seems to me that there's there's an abundance of agreement in this meeting I I don't see uh any big controversy um so uh that's nice um the uh I think what Brynn said uh is it really captures it I liked it because it because what he said was so

experienced near you know he was just describing what it feels like to be a patient and and and and you know to be in a state of and not knowing uh and uncertainty as opposed to you know a more stable kind of uh tranquility and then the the the ability to uh to to allow yourself to return to uncertainty um if you've if you've um found uh answers that are too simple and that that are that are too static to allow ourselves to go back into this state of of not really knowing um that's that I think really does describe what what we try to do for our patients what we have to remember applies to ourselves not only because we are ultimately patients too but because it applies to how the mind works it applies to science how science works you know that we we do need to find order uh but uh there is disorder out there and so we're always at risk of coming prematurely to a rigid system of belief um and we have to return back to the coal face and back to the chaos but ultimately where we want to be is back in in that Valley uh uh um and uh so the as much as Alexis right to remind us that at the uh that the the the the highest levels of brain activity um in many senses of the word uh are entropic um and that that deep sleep and seizures you know are are are bad things as much as that's true nevertheless I think that the great trend of the mind it's it's it's one of its fundamental working principles is where we're wanting to be more certain to we want life to be more predictable we don't like to be in chaos um and I think that that's an important a thing to remember you know while we are acknowledging the importance of of chaos and and of the the highly unpredictable nature of the world and of the phenomena we study in our field nevertheless we Retreat from it and and for very good reasons so those are my closing remarks uh and and thank you and I'll be very brief uh Alexa yeah just thank you for the paper and and uh for uh having such a um strong focus on the patient and on Psychopathic well on on human The Human Condition I I think that's so nice to have that um come out uh of the paper when at at first glance it doesn't seem to be there so thanks for making it so human and then just uh Daniel for organizing and and to uh spend time with Mark and you know but uh it's it's a great pleasure so thank you very much Alexi final words I want to thank everybody and Mark and Brynn and Albert and Daniel Andrea thank you this is I had a great fun and I'm very grateful honestly for your uh help and openness and flexibility one of the other reasons is uh kind of the the time will live in you've all Herrera made that comment that the rate of change is accelerating you know when we were hunter-gatherers uncertainty was Norm you wake up there's nothing in the fridge you have to go hunt and we tolerate an uncertainty better we could put our wounded positions on the back and then we started Living in the world of iPads iPhones were bombarded with information and we must minimize uncertainty we must you know have predictability in everything which

is not the natural world or mammals and now you know we live our life and then vain called it hits and then Bank in a certain person becomes the president of the United States you know and everything goes upside down so uh the black swans are hitting us more frequently so I think having the theory of that you know at our fingerprints and applying guitar work is probably you know beneficial but again thank you all so much I really appreciate awesome well thank you all for joining the conversation continues so till next my birth foreign